

Nikolai Köhler

Curriculum vitae

Junior Research Group Leader

Integrative Food Systems Analysis
Leibniz Institute for Food Systems Biology
at the Technical University of Munich

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Experience

Junior Research Group Leader

Integrative Food Systems Analysis
Leibniz Institute for Food Systems Biology at TUM

Freising, Germany
2026-present

Postdoctoral Researcher

Multifactorial Data Analysis and Machine Learning in the Life Sciences Lab
Heidelberg University

Heidelberg, Germany
2024 - 2026

Postgraduate Fellow

Department of Genetics & Computer Science (Krishnaswamy Lab)
Yale University

New Haven, CT, US
2023

Doctoral Researcher

Chair for Experimental Bioinformatics
Technical University of Munich

Freising, Germany
2020 - 2023

Internship Researcher

Chair for Plant Biochemistry | Metabolomics Australia

Melbourne, Australia
2017 - 2018

Education

Technical University of Munich

Dr. rer. nat. (Bioinformatics), *summa cum laude*
Thesis Title: "Graph-Based Methods for the Analysis and Integration of Lipidome and Metabolome Data into the Omics-Landscape"

Freising, Germany
2020 - 2024

Technical University of Munich

MSc Molecular Biotechnology, *with high distinction*
Thesis Title: "Analysis of Organ-specific Lipidome Compositions and their Network Interactions in Mice"

Freising, Germany
2018 - 2020

Technical University of Munich

BSc Agricultural Science, *with distinction*
Thesis Title: "Regulation of Pyrrolizidine Alkaloid Biosynthesis in *Crassocephalum crepidioides*"

Freising, Germany
2014 - 2018

Publications

- [1] F. Niedermaier, K. Mechteridis, K. Pelz, V. Würf, **N. Köhler**, and J. K. Pauling, Navigating the Lipid Universe with LipidLibrarian: A Cross-Linked Database for Lipidomics Data Integration, *Journal of Lipid Research* 101049 (2026).
- [2] O. I. Coleman[†], A. Sorbie[†], A. Riva, M. von Stern, S. Kuhls, D. M. Selegato, L. Siegert, I. Keidel, **N. Köhler**, J. Wirbel, and others, ATF6 activation alters colonic lipid metabolism causing tumour-associated microbial adaptation, *Nature Metabolism* 1 (2025).
- [3] K. Siebert, T. Faro[†], **N. Köhler**[†], H. Hölz, S. Jarosch, M. Matchado, D. Häcker, F. De Zen, M. S. Hajji, E. Lurz, and others, Endoscopic healing in pediatric IBD perpetuates a persistent signature defined by Th17 cells with molecular and microbial drivers of disease, *Cell Reports Medicine* 6, (2025).

- [4] D. Häcker, K. Siebert, B. J. Smith, **N. Köhler**, A. Riva, A. Mahapatra, H. Heimes, J. Nie, A. Metwaly, H. Hölz, and others, Exclusive enteral nutrition initiates individual protective microbiome changes to induce remission in pediatric Crohn's disease, *Cell Host & Microbe* **32**, 2019 (2024).
- [5] V. Würf, **N. Köhler**, F. Molnar, L. Hahnefeld, R. Gurke, M. Witting, and J. K. Pauling, LipiDetective a deep learning model for the identification of molecular lipid species in tandem mass spectra, *Biorxiv* 2024 (2024).
- [6] T. Damiani, S. Bonciarelli, G. G. Thallinger, **N. Köhler**, C. A. Krettler, A. K. Salihoglu, A. Korf, J. K. Pauling, T. Pluskal, Z. Ni, and others, Software and computational tools for LC-MS-based epilipidomics: challenges and solutions, *Analytical Chemistry* **95**, 287 (2023).
- [7] F. Hoheneder, C. E. Steidele, M. Messerer, K. F. Mayer, **N. Köhler**, C. Wurmser, M. Heß, M. Gigl, C. Dawid, R. Stam, and others, Barley shows reduced Fusarium head blight under drought and modular expression of differentially expressed genes under combined stress, *Journal of Experimental Botany* **74**, 6820 (2023).
- [8] **N. Köhler**, V. Würf, T. D. Rose, and J. K. Pauling, Identification and Multi-Omics Integration of key-Metabolic Reactions from Untargeted Metabolomics, *Biorxiv* 2023 (2023).
- [9] T. D. Rose[†], **N. Köhler**[†], L. Falk, L. Klischat, O. E. Lazareva, and J. K. Pauling, Lipid network and moiety analysis for revealing enzymatic dysregulation and mechanistic alterations from lipidomics data, *Briefings in Bioinformatics* **24**, bbac572 (2023).
- [10] S. Dieckmann, A. Strohmeyer, M. Willershäuser, S. F. Maurer, W. Wurst, S. Marschall, M. H. de Angelis, R. Kühn, A. Worthmann, M. M. Fuh, and others, Susceptibility to diet-induced obesity at thermoneutral conditions is independent of UCP1, *American Journal of Physiology-Endocrinology and Metabolism* **322**, E85 (2022).
- [11] **N. Köhler**, M. Höring, B. Czepukojc, T. D. Rose, C. Buechler, T. Kröhler, J. Haybaeck, G. Liebisch, J. K. Pauling, S. M. Kessler, and others, Kupffer cells are protective in alcoholic steatosis, *Biochimica Et Biophysica Acta (BBA)-Molecular Basis of Disease* **1868**, 166398 (2022).
- [12] T. D. Rose, T. Bechtler, O.-A. Ciora, K. Anh Lilian Le, F. Molnar, **N. Köhler**, J. Baumbach, R. Röttger, and J. K. Pauling, MoSBI: Automated signature mining for molecular stratification and subtyping, *Proceedings of the National Academy of Sciences* **119**, e2118210119 (2022).
- [13] E. M. Haberl, T. S. Weiss, G. Peschel, K. Weigand, **N. Köhler**, J. K. Pauling, J. J. Wenzel, M. Höring, S. Krautbauer, G. Liebisch, and others, Liver lipids of patients with hepatitis B and C and associated hepatocellular carcinoma, *International Journal of Molecular Sciences* **22**, 5297 (2021).
- [14] **N. Köhler**[†], T. D. Rose[†], L. Falk, and J. K. Pauling, Investigating global lipidome alterations with the lipid network explorer, *Metabolites* **11**, 488 (2021).
- [15] S. Schramm, **N. Köhler**, and W. Rozhon, Pyrrolizidine alkaloids: biosynthesis, biological activities and occurrence in crop plants, *Molecules* **24**, 498 (2019).

[†] indicates equal contribution

Talks & Workshops

Talks

- Graph-Based Metabolic Reaction-Centered Multi-Omics Data Integration, 72nd ASMS Annual Conference, 2023
- Reaction-Centered Metabolic Network Analysis, 4th Munich Metabolomics Meeting, 2022
- Lipid network and moiety analysis for revealing enzymatic dysregulation and mechanistic alterations from data, International Conference on Systems Biology, 2022
- Keynote: "Lipid metabolic network analysis for inferring mechanistic changes in enzyme activity 10th Workshop in Lipidomics, 2022
- Network-based Lipidomics Analysis using the Lipid Network Explorer. Virtual Podium Asia Pacific, 2021
- Investigating Global Lipidome Alterations with the Lipid Network Explorer. 1st International Lipidomics Society Conference, 2021

Workshops

- Workshop on Lipidomics Data Analysis. 8th Lipidomics Forum, 2023
- LipiTUM Workshop on Patient Stratification and Lipid Metabolic Network Analysis. 1st International Lipidomics Society Conference, 2021

Honors & Awards

Awards

- Best Presentation Award - Virtual Podiums Asia Pacific

Scholarships

- PROMOS Travel Scholarship - German Academic Exchange Service

Community Service

Journal Reviewing

- Bioinformatics Advances
- Briefings in Bioinformatics
- Inflammatory Bowel Diseases

Event Organization

- Spatial Omics Lunch Seminar @ Heidelberg University (funded by the Chica & Heinz Schaller Foundation)

Teaching & Supervision

Lectures

- Optimal Transport - Theory and Applications (M.Sc. Scientific Computing & M.Sc. Mathematics, 2025, Heidelberg University)
- Data Analysis for the Life Sciences (M.Sc. Biology, 2024, Heidelberg University)
- Data Analysis for RNA sequencing (B.Sc. Biology, 2024-2025, Heidelberg University)
- Computational Biochemistry (M.Sc. Biochemistry, 2020-2022, TUM)

Practicals

- Systems BioMedicine (M.Sc. Bioinformatics, 2020-2022, TUM)
- Problem Based Learning (B.Sc. Bioinformatics, 2021, TUM)

Thesis Supervision

Completed

- Bioinformatics (B.Sc.)
 - “Development of a Deep Learning Model for the Detection and Prediction of Characteristic Fragmentation Patterns in Lipid Mass Spectra”
 - “Network Integration of Metabolome and Microbiome Data using Local Search Optimisation”
 - “Evaluating Linear Model-based Reaction Approximation and Multi-Omics Integration”
 - “Imputing Unmeasured Lipids to Improve the Connectivity of Lipid Metabolic Networks”
- Molecular Biotechnology (M.Sc.)
 - “A Network-based Meta-Analysis to Link Nutritional Metabolites to Lipid Metabolism and Related Diseases”
- Biology (B.Sc.)
 - “Incorporating subcellular localization into dimension reduction algorithms”

Ongoing

- Physics (M.Sc.)
 - “Predicting RNA molecule localization from cellular neighborhood information”

References

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Dr. Josch K. Pauling
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